



# Artificial neural network prediction of transverse modulus in humid conditions for randomly distributed unidirectional fibre reinforced composites: A micromechanics approach

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## ABSTRACT

This paper proposes an innovative micromechanics-based artificial neural network (ANN) method to efficiently investigate the transverse modulus of unidirectional fibre/epoxy composites under humid conditions. In this research, a novel approach is developed to establish relations between the geometrical, mechanical, and environmental properties of the microstructure and the material's performance under transverse tension. A framework is developed to artificially generate periodic representative volume elements (RVEs) while taking into account the interphase region between fibre and matrix. The RVEs are analysed by the finite element method to obtain the transverse moduli of composites. Two-point correlation functions and principal component analysis techniques are applied to extract and compress statistical information from microstructure images. A database establishment framework is developed to create three batches of data with 1000, 2000, and 3000 sizes. An ANN-based prediction framework is developed by integrating 10-fold cross-validation and Bayesian optimisation to optimise the neural network architecture and establish an efficient structure-property linkage under the influence of humid conditions. The prediction results demonstrate the efficiency of ANN in mapping microstructural data to an effective transverse modulus. A parametric study by ANN reveals the role of microstructure geometrical features and humid environmental parameters on the transverse performance of the composite.

## 1. Introduction

In recent years, there has been an increase in demand for lightweight and high-performance materials across various industries, including aerospace, automotive, and construction. Fibre-reinforced polymer (FRP) composites have emerged as promising candidates due to their prominent mechanical properties, such as a high strength-to-weight ratio and fatigue resistance. Although these materials provide promising initial mechanical properties that meet the design requirement, their long-term performance can degrade in interaction with the service environment [1,2]. Humidity is one of the common environments that can impact the durability of composite materials. Previous studies have shown that moisture absorption due to exposure to humid conditions can impact the different mechanical properties of composites. It is seen that moisture can degrade their shear strength, flexural strength, and stiffness [3,4,5]. Moisture can also influence the impact response of the composites [6]. The reason for these modifications comes from the changes induced by water in the constituent materials of composites. A

FRP composite is a heterogeneous material that comprises three phases: fibre, matrix, and interphase. The mechanical properties of FRP are influenced by the properties of its constituents and the geometric arrangements of the phases in the microstructure [7,8]. Once exposed to a humid environment, water creates reversible and irreversible changes to the epoxy matrix that compromise its stiffness [9,10]. The chemical reaction between water and the fibres, particularly the glass fibre, can induce stress corrosion, leading to microcrack growth [11]. Moreover, humidity also weakens the interfacial bond between fibre and matrix, resulting in microcracks in the matrix [12].

Establishing structure-property links in composite materials allows understanding how microstructural factors influence the overall performance of materials. Besides experimental techniques, the composite's mechanical properties can be addressed analytically and by homogenisation techniques such as the rule of mixtures, Mori-Tanaka, and self-consistent methods [13,14,15]. Although these approaches are effective for microstructures with simple geometry, their accuracy drops in relation to complex microstructures [16]. Finite element (FE) methods

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are known as high-fidelity alternatives that accurately capture the relationship between the microstructure geometry and the material's property. However, in the FE method, the representative volume element (RVE) of the composite needs to be large enough to model the mechanical response of the heterogeneous material. This can undermine the computational efficiency of FE, particularly for simulating nonlinear responses or crack propagation in composites with complex microstructural geometries [17,18].

Machine learning has demonstrated remarkable success in the efficient modelling of complex systems and capturing intricate relationships between input and output [19,20,21]. The dependence of mechanical properties on the material's microstructure enables machine learning methods to map the effective properties to the microstructure morphology in an accurate and computationally efficient manner [22,23,24]. Several studies explored the structure–property relationship of different materials by integrating deep learning methods such as convolutional neural networks (CNN) or recurrent neural networks (RNN) with FE analysis [25,26]. In these studies, the microstructural images were artificially generated and used to train the prediction model. Despite its accuracy, the training process of a deep learning model with a large dataset of images is costly and exhaustive [27].

Statistical characterisation of the microstructure morphology by methods such as n-point correlation and lineal path functions is a way to quantify the correlation between different phases in the material [28,29]. However, quantifying the spatial correlations between all pixels in an image result in quite large data. Therefore, dimensionality reduction methods such as principal component analysis (PCA) have been offered to compress the size of the information from microstructure images [30]. Statistical information from the microstructure morphology with compressed size can be used to train machine learning methods more efficiently compared to deep learning models. Several studies have established structure–property linkages by training different prediction models such as support vector machines (SVMs), decision tree ensemble models, and artificial neural networks (ANNs) using two-point correlation (TPC) and PCA functions [31,32,33].

The existing literature lacks a thorough exploration of structure–long-term property relationships in FRP composites exposed to humid environments. This gap hinders our understanding of the degradation of these materials under adverse conditions. Hence, in this study, a novel approach is proposed by integrating an ANN model with micromechanical FE simulations to investigate the degradation in the transverse performance of the unidirectional (UD) glass fibre/epoxy composite under humid conditions. A novel two-dimensional RVE generation framework is developed to automatically create periodic microstructures with random fibre distributions. The generated RVEs are able to capture the periodic nature of the composite microstructures. Periodic boundary conditions (PBCs) are applied to the edges of RVEs to achieve a more realistic representation of the composite material's behaviour. The transverse properties of the UD RVE composites are determined using micromechanical FE simulations. Numerical, analytical, and experimental results are also compared to confirm the validity of the FE modelling. A statistical characterisation technique is used to extract the physical descriptors from the microstructure images using the TPC functions. PCA is employed to reduce the dimension and select important geometric features extracted from the images. The obtained information from the images is integrated with composite constituent properties and humid environmental parameters to establish three batches of training datasets with different sizes. An ANN-based prediction framework is developed by integrating Bayesian optimisation and 10-fold cross-validation (CV) data sampling technique to optimise the neural network architecture. The developed ANN is able to efficiently model the degradation of the composite transverse performance under a variety of humid conditions. The proposed framework in this study suggests a computationally efficient and accurate mapping of geometrical, mechanical, and environmental information of microstructures to the macroscale transverse performance of composites exposed to humid

ageing.

## 2. Generation of representative volume element (RVE)

RVE of a heterogeneous material is identified as the volume that is large enough to represent the mechanical response of the composite [34]. Although different numerical tools have been proposed in the literature to create RVEs [35,36], an automatic generation of a periodic UD micromodel while including an interphase region between fibre and matrix is still lacking.

In this study, a new algorithm based on random sequential adsorption (RSA) method is developed to generate spatially random inclusions with variable sizes in a 2D matrix [37]. In this method, the fibre sizes and locations are randomly and sequentially generated within a pre-defined space. For each fibre, its position is checked with the position of the existing particles in the domain to avoid any overlapping between the fibres. The fibre inclusion in the RVE is continued until the desired volume fraction is achieved. Fig. 1 depicts the flowchart of the RVE generation process which was adopted in this research. In this figure,  $R$ ,  $e$ ,  $L$ ,  $H$ , and  $V_f$  denote the fibre radius, interphase thickness, the RVE length and height, and the desired fibre volume fraction, respectively, that are used as input to the algorithm. An example of an RVE with  $150 \times 150 \mu\text{m}^2$  area,  $0.5 \mu\text{m}$  interphase thickness, and 35 % fibre volume fraction is depicted in Fig. 1.

Since the coordinates of the fibre centres are generated randomly, it is possible that the fibres intersect each other. Therefore, it is necessary to check for overlap before placing them in the modelling space. The distance between a  $k$ -th circle with coordinate  $(X_k, Y_k)$  and the rest of the fibres ( $d_i$ ) within RVE is calculated as:

$$d_i = \sqrt{(X_k - X_i)^2 + (Y_k - Y_i)^2} \quad (1)$$

where  $i = 1, 2, \dots, k-1$ . The  $k$ -th fibre is only added if  $d_i \geq \phi[2(R+e)]$  is satisfied for all the existing fibres.  $\phi > 1$  is a dimensionless factor introduced to prevent the fibres from overlapping each other.

Furthermore, for the fibres that are located at the boundaries, a section of their area is placed outside the RVE. In order to apply periodicity, it is required to replicate the out-of-bounds sections on the direct opposite sides of the RVE. This also avoids the wall-effect of the edges, so that the inclusions can penetrate the boundaries, and RVE is viewed as a sample of a larger area [38]. Fig. 2 shows the method for applying periodicity to the RVE. For a fibre with the coordinate  $(X_k, Y_k)$  in the area  $L \times H$ , three types of positions are possible on the boundaries, which are:

1. Fibre on the left/right edges where  $|X_k - X| < (R+e)$  or  $|X_k - (X+L)| < (R+e)$
2. Fibre on the top/bottom edges where  $|Y_k - Y| < (R+e)$  or  $|Y_k - (Y+H)| < (R+e)$
3. Fibre on the corners by one of the following conditions:
  - Fibre on the left-bottom corner, where  $\sqrt{(X_k - X)^2 + (Y_k - Y)^2} < (R+e)$
  - Fibre on the right-bottom corner, where  $\sqrt{(X_k - (X+W))^2 + (Y_k - Y)^2} < (R+e)$
  - Fibre on the left-up corner, where  $\sqrt{(X_k - X)^2 + (Y_k - (Y-H))^2} < (R+e)$
  - Fibre on the right-up corner, where  $\sqrt{(X_k - (X+W))^2 + (Y_k - (Y-H))^2} < (R+e)$

It is seen in Fig. 2 that *Circle-1* is located on the bottom edge. The periodicity of the material requires to replicate the circle on the direct opposite edge. Therefore, *Circle-1<sup>y</sup>* is created on the top edge by translating a copy of the *Circle-1* along the y-axis by  $L$ . A similar

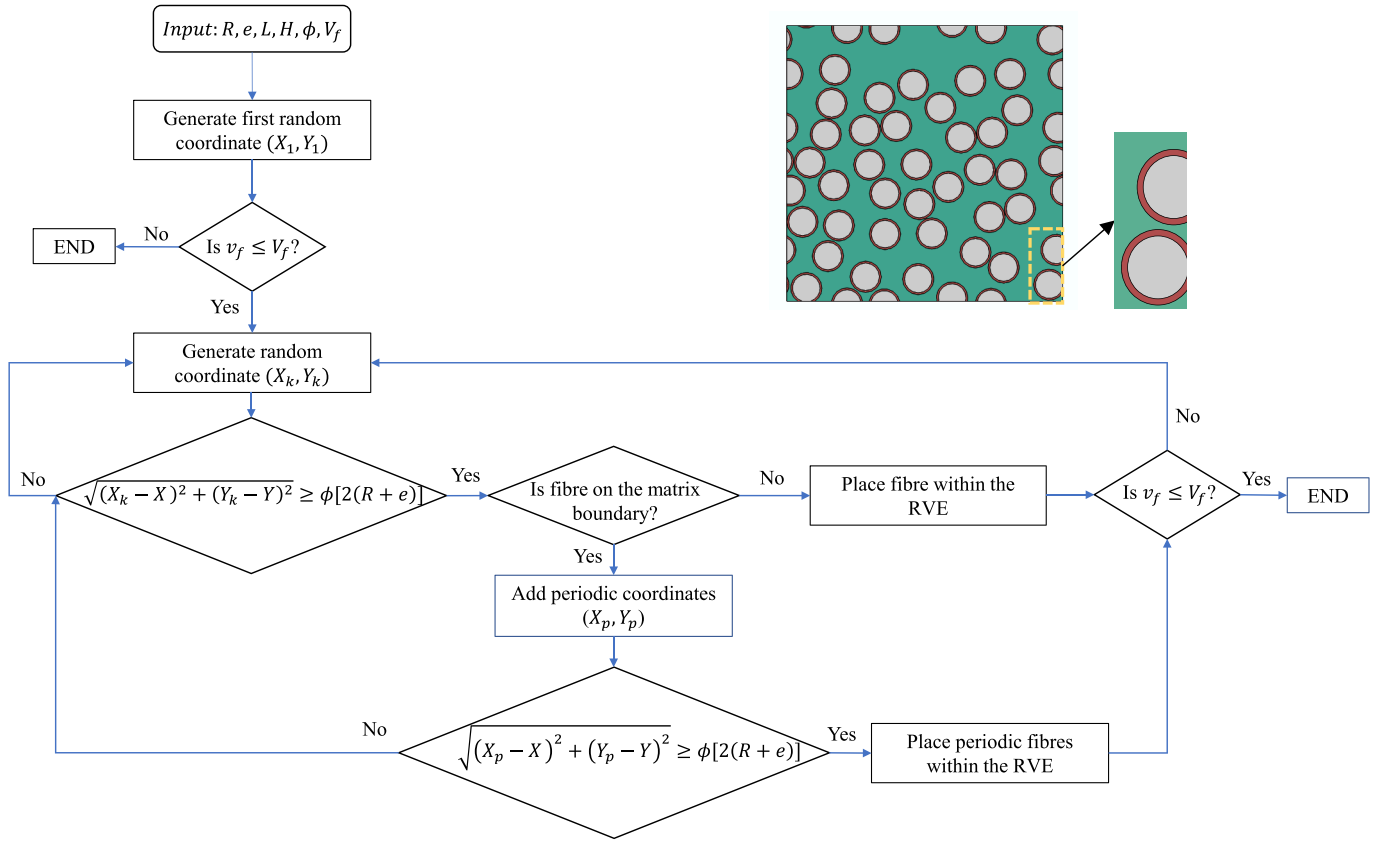


Fig. 1. Flowchart for RVE automatic generation with an example of a created microstructure.

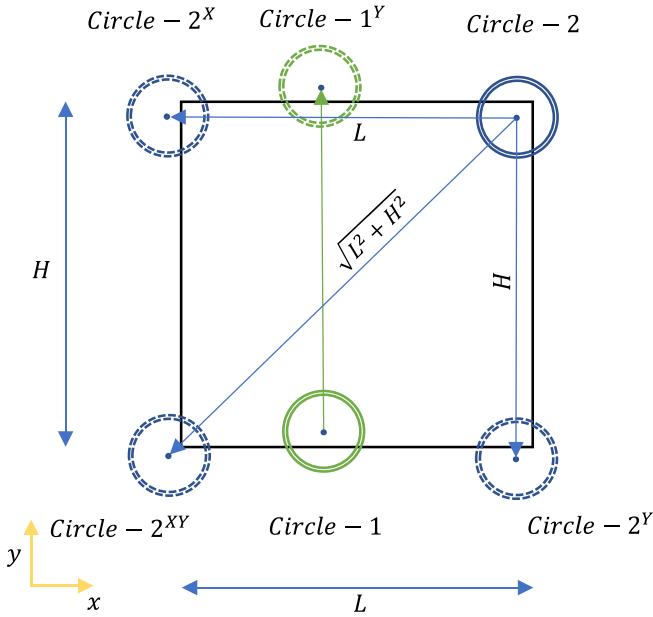


Fig. 2. Applying periodicity of material to the RVE.

transition can be applied to the fibres on the left/right boundaries. In Fig. 2, Circle-2 is located on the right-up corner of the area. It is necessary to duplicate the circle along the x- and y-axes and the diagonal of the area. As a result, Circle - 2<sup>x</sup> is formed along the x-axis by translating a copy of the Circle-2 by  $L$ . Circle - 2<sup>y</sup> is a duplication of Circle-2 along the y-axis by  $H$ , and Circle - 2<sup>xy</sup> is created by moving the centre of a copy of Circle-2 by  $L$  and  $H$  along the x- and y-axes, respectively. The created

circles on the edges and corners are depicted by the dashed lines in Fig. 2.

In order to generate the 2D RVE, a Python script was developed based on the flowchart in Fig. 1 and passed to the Abaqus CAE software [39]. The code ran until the fibre volume fraction ( $V_f$ ) reached a desired value.

### 3. Digital representation of composite material

A composite microstructure usually contains a series of geometric descriptors such as size, volume fraction, arrangement, and relative location of phases. The statistical distribution of these features characterises the composite internal structure. Among the various statistical characterisation methods, n-point spatial correlations have been shown to precisely quantify the geometry of the composite with random inclusions [40,41]. Two-point correlation (TPC) functions have been successfully used in the accurate property prediction and reconstruction of composite materials [7,31]. The TPC function  $f_r^{np}$  quantifies the probability of finding the local states of  $n$  and  $p$  at the start and end of a vector of length  $r$ , randomly placed in a microstructure RVE image with  $S$  pixels. It is written as:

$$f_r^{np} = \frac{1}{S} \sum_{s=1}^S m_s^n m_{s+r}^p \quad (2)$$

where  $s$  is the spatial position of the pixel. For a vector of length  $r$  with its tail located at pixel  $s$ ,  $m_s^n$  is equal to 1 if the material phase  $n$  is present at  $s$  and 0 if any other phase is located in this spatial position. With the assumption of periodic RVE,  $S$  represents the total number of placements for the vector of length  $r$  in the microstructure. For  $n = p$ , Eq. (2) is described as auto-correlation, which states the probability of finding a specific phase  $p$  at the material. However,  $n \neq p$  denotes a cross-correlation function for Eq. (2).

For a three-phase composite material including interphase region, nine TPC functions are defined as:

$$f_r^{np} = \begin{bmatrix} f_r^{FF} & f_r^{FM} & f_r^{FI} \\ f_r^{MF} & f_r^{MM} & f_r^{MI} \\ f_r^{IF} & f_r^{IM} & f_r^{II} \end{bmatrix} \quad (3)$$

where  $F$ ,  $M$ , and  $I$  denote the fibre, matrix, and interphase phases, respectively. Considering that the resulting  $f_r^{np}$  in Eq. (3) comprises a very large dataset, various techniques in the literature have investigated the interdependencies between the TPCs. A study by Niezgoda et al. [40] showed that for a periodic microstructure of  $N$  phases,  $N-1$  correlations are independent. Therefore, following this research, calculating only two of the nine TPC functions in Eq. (3) are sufficient to obtain the remaining functions in a three-phase material. Considering the convolution nature of Eq. (2), discrete Fourier transforms (DFTs) were used to accelerate the calculations of the TPC functions [42].

Fig. 3 shows an example of a microstructure with a 30 % fibre volume fraction and its corresponding TPC plots for  $f_r^{FF}$ ,  $f_r^{FM}$ , and  $f_r^{FI}$ , respectively. The centre of the contour in Fig. 3b corresponds to  $r = 0$  and has an intensity value of 0.3 which is equal to the fibre volume fraction. Fig. 3c illustrates the cross-correlation function between fibre and matrix. This contour implies the probability of finding a vector of length  $r$ , starting at the fibre's centre and ending at the matrix phase. Similarly, Fig. 3d depicts the intensity values for the vectors with the head and tail in interphase and fibre phases, respectively. These contours provide useful information regarding the volume fraction and the ratio between the fibre diameter and the RVE length [31].

#### 4. Principal component analysis

PCA is a data compression technique that has been vastly used in machine learning applications [31]. It is done by projecting the  $n$ -dimensional data onto a space with  $k$  orthogonal principal axes, where for each axis, the variance of the projected data is maximum. These axes

are numbered in descending variance order. In this case, the first principal component (PC<sub>1</sub>) is derived by decomposing the original data along the eigenvector with a direction that results in the highest variance in the projected data. Subsequently, the following principal components (PCs) are defined as the eigenvectors that are orthogonal to the preceding ones and the variances of the data along them are in a descending order [43]. Once the eigenvectors are derived, the data are projected into the space with new principal axes.

The mathematical principle of the PCA is based on the singular value decomposition of the data matrix  $A$  with the size  $n \times p$ , which contains the TPCs as:

$$A = \begin{bmatrix} f_{r_{n=1,p=1}}^{FF} & \cdots & f_{r_{n=1,p=649,800}}^{MM} \\ \vdots & \ddots & \vdots \\ f_{r_{n=649,800,p=1}}^{FF} & \cdots & f_{r_{n=649,800,p=649,800}}^{MM} \end{bmatrix} = USV^T \quad (4)$$

In Eq. (4),  $S$  is a  $p \times p$  diagonal matrix with the eigenvalues of the principal components.  $U$  is a  $n \times p$  orthonormal matrix with the columns representing the principal axes of  $A$ . Also,  $V$  is a  $p \times p$  matrix which contains the orthogonal eigenvectors of the matrix  $A^T$ . Once the matrix  $A$  is decomposed into matrices of eigen values and eigenvectors, dimensionality reduction is performed by reconstructing the matrix  $A$  using the first  $k$  principal components, where  $k < \max(n,p)$ . The new matrix  $A'$  is written in a reduced dimension as:

$$A'_{(n \times k)} = U_{(n \times k)} S_{(k \times k)} V_{(k \times k)}^T \quad (5)$$

The principal components are ranked based on their corresponding eigenvalues. Therefore, the first component has the greatest variance. The presence of redundant features in the data can compromise the prediction model's performance as they cause overfitting and increase the complexity of the model [44].

#### 5. FE simulation and validation

While fibres offer high strength in the longitudinal direction of a UD

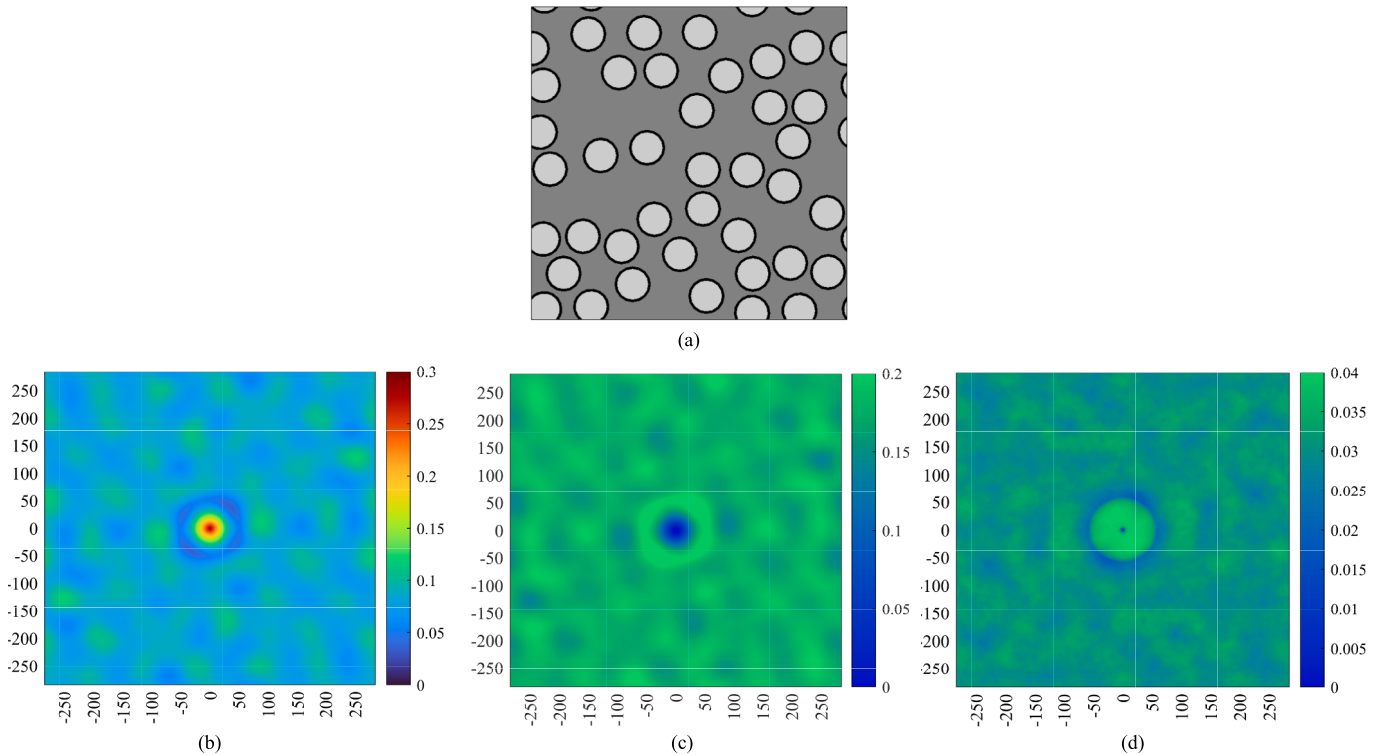


Fig. 3. Top: 2D UD composite microstructure with 30% fibre volume fraction; Bottom: TPC function contour plots: (b)  $f_r^{FF}$ , (c)  $f_r^{FM}$ , (d)  $f_r^{FI}$ .

composite, their mechanical properties in the transverse direction are limited. This limitation results in low strength in transverse direction and stress concentration at the fibre and matrix interface, leading to micro-cracking and eventually fracture of the material [17]. Therefore, it is necessary to investigate the performance of the composites in the transverse direction. FE method is used to simulate the transverse behaviour of the 2D RVE while incorporating the geometry, material property and boundary conditions of the composite material's constituents. The UD composite components, namely, fibre, matrix, and interphase are considered homogenous and linear materials with isotropic mechanical behaviour and plain strains. In addition, the interface is assumed to be perfectly tied to the fibre and matrix [16,45].

Periodic boundary conditions (PBCs) are applied as a set of constraints to the opposing edges of the RVE. PBC implies that the shape of the opposite edges remains identical under a particular loading history [38]. Also, to maintain stress continuity across the borders, the stress vectors on the opposite edges are in the opposite directions. The use of PBC has demonstrated its capacity to provide accurate estimations of the effective mechanical properties [46,47]. Fig. 4 shows a 2D RVE of circular fibres where  $N_L$ ,  $N_R$ ,  $N_T$ , and  $N_B$  represent the nodes on left, right, top, and bottom edges, respectively. In order to obtain the RVE transverse behaviour, it is necessary to place certain constraints on the nodes  $N_1$ ,  $N_2$ , and  $N_4$  in Fig. 4. In this figure,  $N_1$  is fixed along  $x$ - and  $y$ -axis to prevent rigid body motion. A tensile uniaxial loading parallel to  $x$ -axis is applied to  $N_4$  by defining a positive displacement  $\delta_x$ , while the node  $N_2$  is constraint along the  $x$ -axis. The periodicity of displacement requires that the nodes on the opposing edges are kinematically constrained. Therefore, for the representative nodes ( $N_L$ ,  $N_R$ ,  $N_T$ , and  $N_B$ ) on the edges of the RVE, the mathematical formulation of PBC is written as:

$$U_{(x,y)}^{N_R} = U_{(x,y)}^{N_L} + U_{(x,y)}^{N_4} - U_{(x,y)}^{N_1} = 0 \quad (6)$$

$$U_{(x,y)}^{N_T} = U_{(x,y)}^{N_D} + U_{(x,y)}^{N_2} - U_{(x,y)}^{N_1} = 0 \quad (7)$$

where  $U_{(x,y)}^{N_i}$  explains the displacement of the node  $N_i$  in  $x$ - and  $y$ -axes. Eq. (6) constraints the displacement of the node  $N_R$  to the degrees of freedom in nodes  $N_L$ ,  $N_4$ , and  $N_1$ . Similarly,  $U_{(x,y)}^{N_T}$  is related to the displacements in nodes  $N_D$ ,  $N_2$ , and  $N_1$  in Eq. (9). These equations are the basis for the mathematical application of PBCs to the RVE.

It is assumed that the difference between the RVE and the macro-model length scales is sufficiently large so that the principle of the separation scales can be applied [48]. Considering the first-order homogenisation and zero boundary fluctuation model [49], the macro

strain  $\epsilon_{macro}$  is applied to the boundary of the RVE, thereby the displacement vector is written as:

$$\delta_x = \epsilon_{macro} \Delta x \quad (8)$$

where  $\Delta x$  is the relative position vector. The macroscale stress tensor is also computed following the Hill-Mandel principle as a volume-averaged stress at the microscale, which gives:

$$\sigma_{macro} = \frac{1}{\Gamma} \int_{\Gamma_{RVE}} \sigma_{micro} d\Gamma \quad (9)$$

where  $\Gamma$  is the RVE domain. The average stress and strain are related with the stiffness tensor  $C$  in the form of:

$$\sigma_{macro} = C \epsilon_{macro} \quad (10)$$

By choosing a unit strain value, the components of the  $C$  matrix are obtained by averaging the stress tensor over the elements.

The validity of FE simulations was checked by the existing experimental measurements of transverse performance in glass fibre-reinforced epoxy composites with different constituent properties and under a variety of humid conditions. Table 1 shows the unaged and degraded properties of the composite constituents impacted by a range of relative humidity (RH), temperature, and duration parameters. For the fibre, matrix, and interphase, in this table, the elastic modulus is denoted by  $E_m$ ,  $E_f$ , and  $E_i$ . Similarly, the Poisson ratio of these materials is shown by  $\nu_m$ ,  $\nu_f$ , and  $\nu_i$ , respectively. The thermal expansion coefficients for fibre ( $\alpha_f$ ) and matrix ( $\alpha_m$ ) are reported in Table 1 for different references. Also, the moisture expansion coefficient of the matrix ( $\alpha_w$ ) is shown in this table. The fibre volume fractions of the composite samples tested by Riaño et al. [16], Joliff et al. [50], Naya et al. [18], and Rocha et al. [51] are 20 %, 50 %, 60 %, and 60 %, respectively. It is known that the jamming limit of the RSA algorithm prevents it from generating RVEs with fibre volume fractions greater than 55 % [52]. The proposed method for generating RVEs in this study was used to create fibres with diameters having a mean distribution of 13.2  $\mu m$  and a standard deviation of 0.8  $\mu m$  for the composites with 60 %  $V_f$ . The elastic properties of the interphase region were reported by Riaño et al. [16] and Joliff et al. [50] were measured using atomic force microscopy (AFM). The evolution of effective Young's modulus within the distance from the fibre surface revealed a decrease in properties until reaching a plateau within the range of 500 nm to 1  $\mu m$ , which is regarded as the thickness of the interphase area.

The information on the constituent properties was used within the FE simulation to estimate the degraded transverse modulus of the composite materials. The simulations were conducted using Abaqus CAE, with the RVEs meshed by the CPE4 elements and a size of  $1.5 \times 1.5 \mu m^2$ . In order to select an optimal RVE size, a parametric study was conducted to investigate the change in the Young's modulus for the models with 20 % fibre volume fraction and different RVE length to fibre diameter ( $L_{RVE}/D_{fibre}$ ) ratios. The results are presented in Fig. 5, and it is seen that consistent results are obtained for  $L_{RVE}/D_{fibre} \geq 10$ .

Fig. 6 depicts the scatterplot of the environmental parameters as well as the stress contours of composite materials at the aged and unaged stages for various references. Fig. 6a–h illustrates the relevant stress plot counterpart for each point on the scatter plot. The von Mises stress isolines of aged composite materials are shown in Fig. 6e–h, respectively. The corresponding unaged stress plot for each aged model is also shown in Fig. 6a–d, respectively. To emphasis the impact of ageing and avoid displaying stress levels that are too high, the stress intensities were normalised to those of unaged models. This is due to the fact that no damage initiation criteria were considered in the modelling. As a result, the stress values pass the glass fibre composite ultimate stress in the transverse direction, which is 30–40 MPa [53].

The comparison of stress levels between the aged RVEs and their corresponding unaged micromodels in Fig. 6 reveals that unconditioned

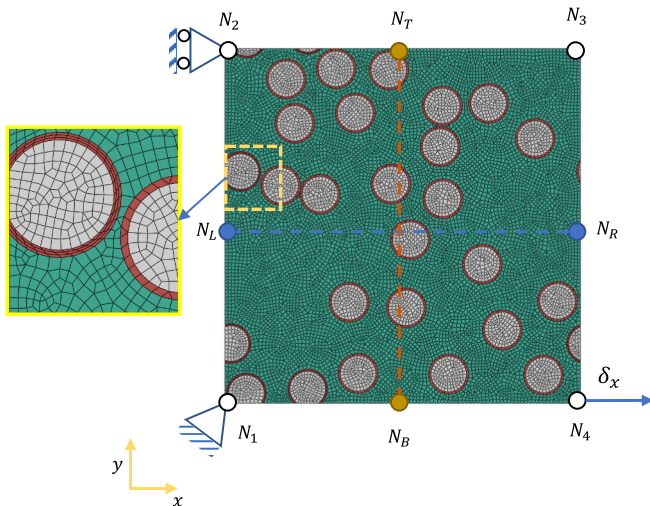
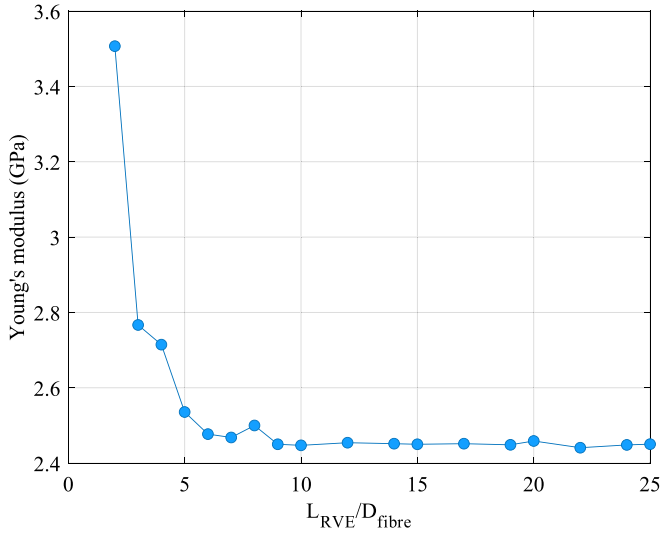


Fig. 4. Numerical model of a periodic 2D RVE meshed with CPE4 elements and under uniaxial tensile displacement.

**Table 1**

Details of the experimental studies on the transverse performance of composite materials under humid conditions.

| Reference          | Environmental properties |                  |        | Mechanical properties |                |                                   |                                   |                      |                |                                   |                      |                |
|--------------------|--------------------------|------------------|--------|-----------------------|----------------|-----------------------------------|-----------------------------------|----------------------|----------------|-----------------------------------|----------------------|----------------|
|                    | Duration (weeks)         | Temperature (°C) | RH (%) | Matrix                |                |                                   |                                   | Fibre                |                |                                   | Interphase           |                |
|                    |                          |                  |        | E <sub>m</sub> (GPa)  | ν <sub>m</sub> | α <sub>m</sub> (K <sup>-1</sup> ) | α <sub>w</sub> (% <sup>-1</sup> ) | E <sub>f</sub> (GPa) | ν <sub>f</sub> | α <sub>f</sub> (K <sup>-1</sup> ) | E <sub>i</sub> (GPa) | ν <sub>i</sub> |
| Riaño et al. [16]  | 0                        | 25               | 50     | 1.43                  | 0.4            | –                                 | –                                 | 72                   | 0.25           | –                                 | 1.9                  | 0.4            |
|                    | 76                       | 25               | 80     | 1.2                   | 0.4            | –                                 | –                                 | 72                   | 0.25           | –                                 | 1.1                  | 0.4            |
| Joliff et al. [50] | 0                        | 25               | 50     | 2.7                   | 0.35           | –                                 | 0.001                             | 70                   | 0.35           | –                                 | 2.7                  | 0.35           |
|                    | 16                       | 70               | 100    | 2.5                   | 0.35           | –                                 | –                                 | 70                   | 0.35           | –                                 | 2.5                  | 0.35           |
| Naya et al. [18]   | 0                        | 25               | 50     | 5.07                  | 0.35           | 52e-6                             | –                                 | 12.97                | 0.3            | 7.2e-6                            | –                    | –              |
|                    | 3                        | 70               | 85     | 4.28                  | 0.35           | 1.5e-6                            | –                                 | 12.97                | 0.3            | 7.2e-6                            | –                    | –              |
| Rocha et al. [51]  | 0                        | 25               | 50     | 3.13                  | 0.37           | 6e-5                              | 0.002                             | 73                   | 0.22           | 5.4e-6                            | –                    | –              |
|                    | 4.5                      | 50               | 100    | 2.50                  | 0.37           | –                                 | 0.002                             | 73                   | 0.22           | 5.4e-6                            | –                    | –              |

**Fig. 5.** Influence of  $L_{RVE}/D_{fibre}$  on Young's modulus.

samples exhibit greater stress values. This observation is related to the comparatively greater stiffness of the matrix during the early stages as opposed to the aged model. In addition, the increase in size and drop in the mechanical properties of the interphase region due to exposure to high humidity over time contribute to the decrease in stress levels seen in the aged model [16].

A comparison between the FE, experimental, and analytical results was conducted to confirm the validity of the FE simulations. Two analytical models called Mori-Tanaka [14] and the Hashin and Shtrikman [54] were used to obtain the effective transverse modulus. In the Mori-Tanaka (M-T) model, the FRP composite effective stiffness tensor is in the form of:

$$E_{eff} = E_m + [\nu_f \langle (E_f - E_m) A_{Eshelby} \rangle] [\nu_m I + \nu_m \langle A_{Eshelby} \rangle]^{-1} \quad (11)$$

where  $\langle \bullet \rangle$  denotes the fibre orientation averaging, and  $A_{Eshelby}$  is the strain localisation tensor of the dilute solution which is written as:

$$A_{Eshelby} = [I + SE_m^{-1}(E_f - E_m)]^{-1} \quad (12)$$

In Eq. (10),  $I$  is the identity tensor, and  $S$  denotes the Eshelby elastic tensor [55].

For the Hashin and Shtrikman, the lower bound for elastic modulus ( $E^{HS-}$ ) is calculated as:

$$E^{HS-} = \frac{9K^- G^-}{3K^- + G^-} \quad (13)$$

where

$$K^- = K_m + \frac{\nu_f}{\frac{1}{K_f - K_m} + \frac{3\nu_m}{3K_m + 4G_m}}$$

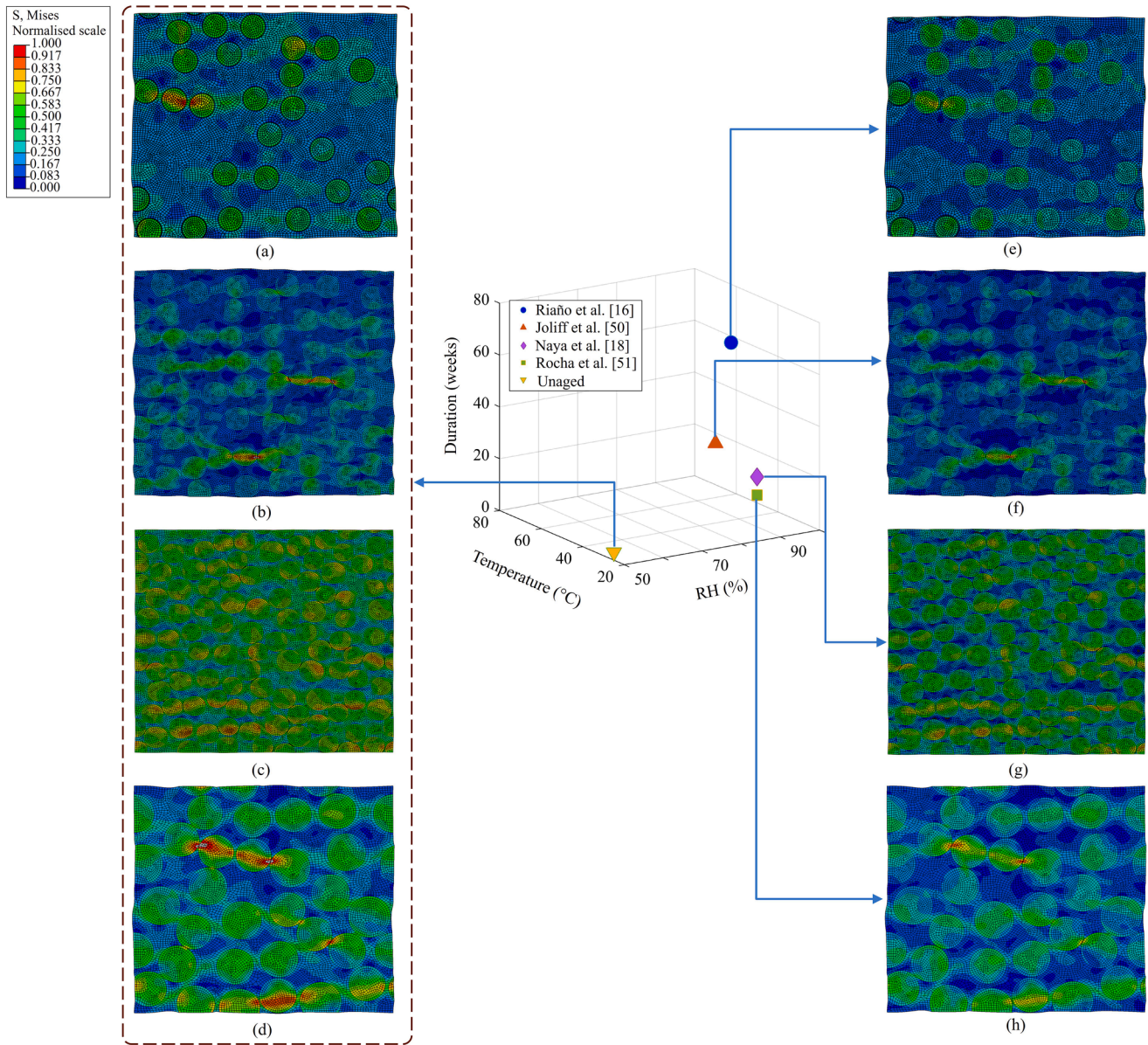
$$G^- = G_m + \frac{\nu_f}{\frac{1}{G_f - G_m} + \frac{6(K_m + 2G_m)\nu_m}{5G_m(3K_m + 4G_m)}} \quad (14)$$

In Eq. (14),  $K_m$  and  $G_m$  denote the matrix bulk and shear moduli, respectively. Similarly,  $K_f$  and  $G_f$  are the corresponding moduli values for the fibre. The values of transverse modulus derived from different methods are presented in Table 2. For the M-T method, the HomoPy Python package, developed for calculating the effective stiffness properties, was used [56]. The difference between the experimental results and the FE and analytical estimations are listed in Table 2. Comparing the difference values indicates that FE estimations are in closer agreement with the experimental measurements than the analytical predictions. This verifies the validity of the FE simulations for predicting material degradation under a variety of environmental parameters. It is seen that the findings of Riaño et al. [16], Joliff et al. [50], Naya et al. [18], and Rocha et al. [51] deviate from FE estimations by 0.0 %, 1.9 %, 23.2 %, and 1.6 %, respectively. The discrepancy values are due to the assumption of a perfect interface tie between the fibre and matrix in the FE simulation. However, it has been shown that moisture weakens the interface between the fibre and matrix, resulting in fibre-matrix debonding [57].

## 6. Machine learning framework establishment

In this study, the artificial neural networks (ANNs) method was used to capture the relationship between the microstructure's geometric, mechanical, and environmental properties and the composite's transverse performance under humid ageing. ANN has become the most widely used model in theory and practice owing to its capability to manage big-data analytics [58]. ANN can represent both complex and simple models, and its calculation cost can be adjusted to meet the problem requirements [59]. An ANN architecture comprises a series of neurons connected with associated weights. In each training step of ANN, feedforward and backpropagation processes are performed to update the weights and reduce the error between the predicted and real output values. Once trained, the ANN model is assigned the optimal weights to calculate the output with the minimum error. Fig. 7 depicts the architecture of ANN with the input, output, and multiple hidden layers iterated in feedforward and backpropagation to predict the transverse elastic modulus ( $\hat{E}_{22}$ ). In this figure,  $x$ ,  $w$ , and  $a$  denote the input feature, the associated weight to the neuron, and the output of the neuron, respectively. For the  $j$ th neuron in the  $k$ th hidden layer, the output is written as:

$$a_j^{(k)} = \sigma \left( \sum_{i=0}^n w_{ji}^{[k]} a_i^{[k-1]} \right) \quad (15)$$



**Fig. 6.** Scatterplot of environmental parameters with the corresponding von Mises stress contours of unaged and aged samples: (a, e) Riaño et al. [16]; (b, f) Joliff et al. [50]; (c, g) Naya et al. [18]; (d, h) Rocha et al. [51].

**Table 2**

Effective Young's modulus of aged micromodels obtained from experimental, analytical, and FE studies.

| Reference          | Experimental result (GPa) | Analytical result (GPa) |                 | FE result (GPa) | Difference (%) |                 |     |
|--------------------|---------------------------|-------------------------|-----------------|-----------------|----------------|-----------------|-----|
|                    |                           | M-T                     | HS <sup>+</sup> |                 | M-T            | HS <sup>+</sup> | FE  |
| Riaño et al. [16]  | 2.06                      | 3.1                     | 2.1             | 2.06            | 32.2           | 2.8             | 0.0 |
| Joliff et al. [50] | 7.7 ± 0.2                 | 7.7                     | 7.0             | 7.7             | 1.9            | 12.8            | 1.9 |
| Naya et al. [18]   | 8.4                       | 10.8                    | 8.0             | 8.7             | 28.6           | 4.8             | 3.5 |
| Rocha et al. [51]  | 11.7 ± 0.4                | 12.5                    | 11.2            | 12.3            | 3.2            | 8.0             | 1.6 |

where  $\sigma(\bullet)$  denotes the activation function which operates on the weighted sum of the inputs to the neuron and determines the neuron's output. The framework for the ANN deployment is also shown in Fig. 7.

A 10-fold cross-validation (CV) with Bayesian optimisation is used to search for the architecture with minimum prediction error. Tuning begins with training ANN using initial hyperparameter values. The validation folds error is then used as a metric for the Bayesian algorithm to search for a new set of hyperparameter values. The tuning process continues until a minimum CV error is achieved for the ANN, ensuring that the model's performance is optimised and robustly validated across validation subsets of the dataset [1].

## 7. Results and discussion

### 7.1. Database establishment

Since machine learning is a data-driven approach, it is first required to establish a database to train the ANN model. It is done by generating and pre-processing microstructure data for predicting the long-term transverse performance of the UD composite. In order to train ANN, first a dataset batch of 3000 size was generated. The created data contained information on the constituent mechanical properties of

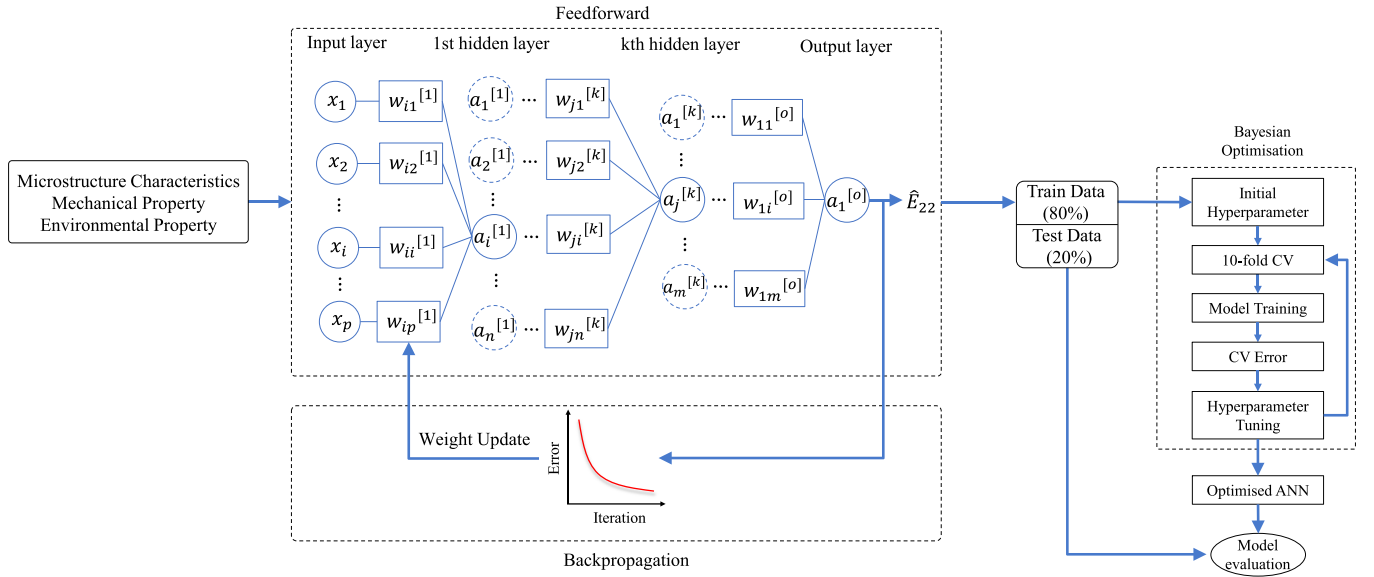


Fig. 7. Flowchart of ANN optimisation and validation.

composites and the humid environmental conditions obtained from research references, namely Riaño et al. [16], Joliff et al. [50], Naya et al. [18], and Rocha et al. [51]. For this batch, RVEs with random fibre distributions and volume fractions ranging from 10 % to 60 % were generated using the framework in Fig. 1. The RVEs were then assigned the values corresponding to the mechanical properties of fibre, matrix, and interphase, as provided in Table 1. In order to check the influence of the training data size on the predictions of the ANN, two other batches were created by randomly selecting 1000 and 2000 data points from the original data batch, allowing for a comprehensive examination of how training data size impacts the performance and generalisation of the ANN.

Each created RVE composite contains three phases, namely fibre, matrix, and interphase, and was saved as an image with  $570 \times 570$  pixels of resolution, resulting in a total of 324,900 pixels. The geometrical information, such as the relative size of each phase with respect to RVE's size, volume fraction, and distribution, was extracted by applying TPC functions to each pixel of the microstructural image. The resulting matrix contained 649,800 two-point spatial correlations for every image. Developing an ANN model using the TPCs from all of the generated RVEs is a computationally expensive task. Therefore, the number of parameters was reduced by selecting the most important features using PCA to improve the efficiency of the prediction model [60]. As a result, the size of the extracted TPC functions of each image was reduced to 20 principal components. In order to quantify the amount of variance captured by each component, the cumulative variance of the PCs is calculated. Fig. 8 shows the change in the variance of the data with the PC number. It is seen that the first two PCs capture about 98.5 % of the variance in the data, indicating that they contain the most prominent geometrical features of the microstructure.

To train ANN, each generated RVE was modelled using FE simulation and labelled by its corresponding transverse modulus ( $E_{22}$ ). For all the FE models, the fibre, matrix, and interphase region were considered homogenous and linear materials with isotropic mechanical behaviour and plain strains [16,45,50]. The geometry of the RVEs were meshed with CPE4 elements of  $1.5 \times 1.5 \mu\text{m}^2$  size to avoid convergence problems, and the PBCs were applied to the edges of the RVE. Fig. 9 depicts the process of establishing a database for ANN model training. It is seen that the degraded mechanical properties for fibre, matrix, and interphase and their corresponding information on duration, RH, and temperature were collected from the four references. For each reference, 750 RVEs with varying fibre volume fraction, fibre arrangements, and

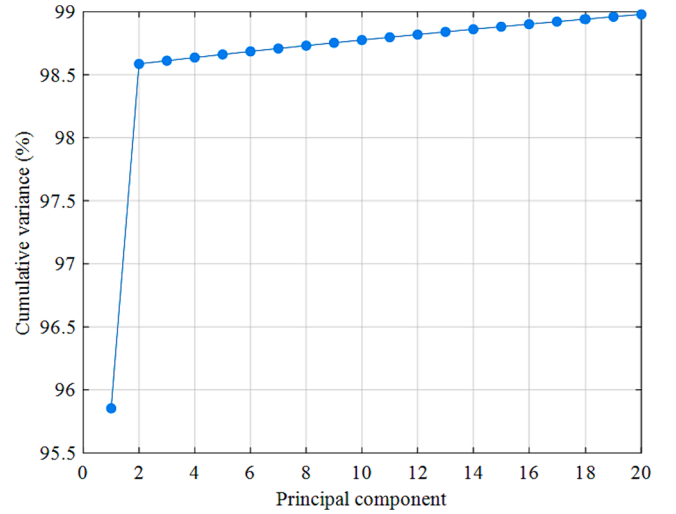


Fig. 8. Cumulative variance of input data for different PCs.

interphase thickness were created. These RVEs were then assigned the mechanical properties of the reference and modelled within the FE simulation to estimate  $E_{22}$ . Table 3 presents the list of the input features in the database with their corresponding ranges. It is seen that the established data comprises information related to the microstructure characteristics, as well as the mechanical and environmental properties of the FRP composites.

## 7.2. ANN model development and validation

In order to develop a prediction model, the created data batches were randomly divided into 80 % training and 20 % test subsets. The training data was cross-validated by 10 different folds to confirm the model's generalisation ability and avoid overfitting. In order to optimise the prediction performance of the ANN, Bayesian optimisation was used to automatically tune the model hyperparameters and minimise its prediction error [1].

Table 4 presents the list of hyperparameters, search ranges, and optimal values obtained by using Bayesian optimisation for the ANN models trained on the three batches. In this table,  $\lambda$  denotes the

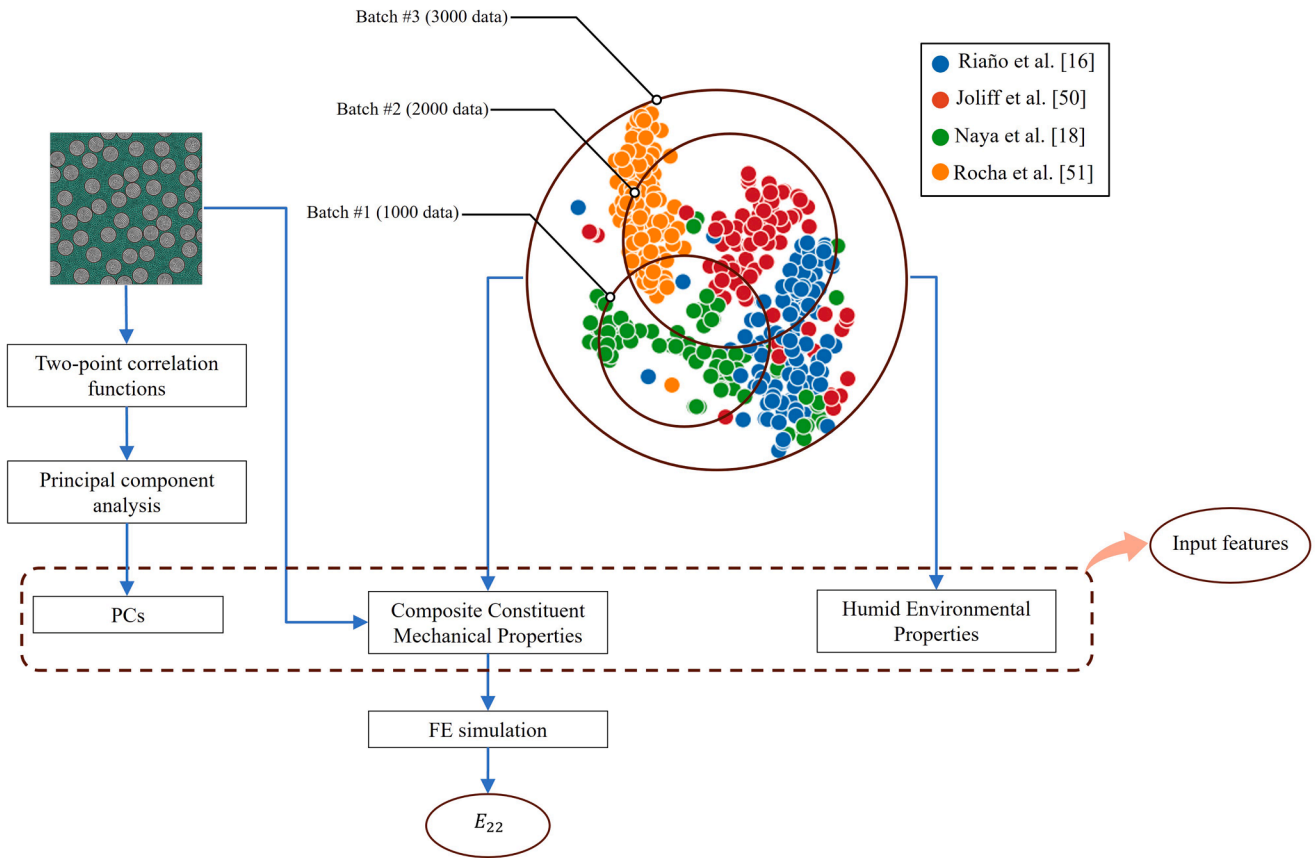


Fig. 9. Flowchart for establishing 3 batches of datasets for ANN model training.

Table 3

Details of the features in the generated database.

| Feature type     | Category                       | Description | Unit  | Range               |
|------------------|--------------------------------|-------------|-------|---------------------|
| Input parameter  | Microstructure characteristics | PC          | –     | (–165)–169          |
|                  |                                | $e$         | nm    | 0–1000              |
|                  | Mechanical property            | $\nu_f$     | %     | 10–60               |
|                  |                                | $E_f$       | GPa   | [12.97, 70, 72, 73] |
|                  |                                | $E_m$       | GPa   | [1.2, 2.5, 4.28]    |
|                  | Environmental property         | $E_i$       | GPa   | [1.1, 2.5]          |
|                  |                                | $RH$        | %     | [80, 85, 100]       |
| Output parameter | Mechanical property            | Temperature | °C    | [25, 50, 70]        |
|                  |                                | Duration    | weeks | [3, 4.5, 16, 76]    |
|                  |                                | $E_{22}$    | GPa   | 1.7–12.4            |

Table 4

List of hyperparameters with selected values for ANN.

| Hyperparameter      | Search range        | Optimal value |           |           |
|---------------------|---------------------|---------------|-----------|-----------|
|                     |                     | Batch #1      | Batch #2  | Batch #3  |
| Number of layers    | 1–3                 | 2             | 2         | 1         |
| Layer size          | 1–300               | 9, 1          | 60, 17    | 15        |
| Activation function | Relu, Tanh, Sigmoid | Sigmoid       | Relu      | Tanh      |
| $\lambda$           | 1.04e(-8) – 104.16  | 4.04E(-8)     | 5.02E(-5) | 4.71E(-6) |
| Weight initialiser  | Glorot, He          | He            | Glorot    | He        |
| Standardisation     | Yes, No             | Yes           | No        | Yes       |

generalisation term used in training, and weight initialiser describes the functions used to sample the weights from either a uniform or normal distribution by Glorot or He functions, respectively [61,62]. The range of the input features in Table 3 showed that the data had different scales for various parameters. Features with different scales can compromise the ANN's prediction ability since the effect of variables with small scales is masked by the ones with greater scales [63]. This necessitates considering the influence of scaling the range and distribution of input variables as one of the hyperparameters in ANN tuning. In Table 4, standardisation denotes a method for scaling the input parameters to have a zero mean and unit variance value. The tuning process of the hyperparameters for the models trained on various data batches is shown in Fig. 10. Each point in this figure shows the minimum of the estimated cost function for the ANN model after it was trained using the hyperparameters obtained in each iteration. The best point implies the set of hyperparameters from which the minimum prediction error is derived. It is seen that minimum error values of 0.1407, 0.0069, and 0.006 were obtained at the 25th, 26th, and 27th iterations of the Bayesian optimisation for the models trained by the first, second, and third data batches, respectively. The learning curves of the ANN models during training with the selected hyperparameters are depicted in Fig. 9. The observed trend in the model trained by the first batch shows a notable decrease in the mean squared error (MSE) followed by a steady convergence towards minimum value within the first 100 epochs. For the ANNs developed by the second and third training data batches, a steep decrease occurs in the first 10 epochs. The difference between the converged errors of the models in Fig. 11 implies that a smaller size of training data in the first batch compromises the accuracy of the developed ANN model compared to the ones trained by the larger datasets.

The prediction performance of the ANN models was checked by introducing evaluation criteria. Two metrics, namely, root-mean-square error (RMSE) and goodness of fit (R2) were used to show the efficiency

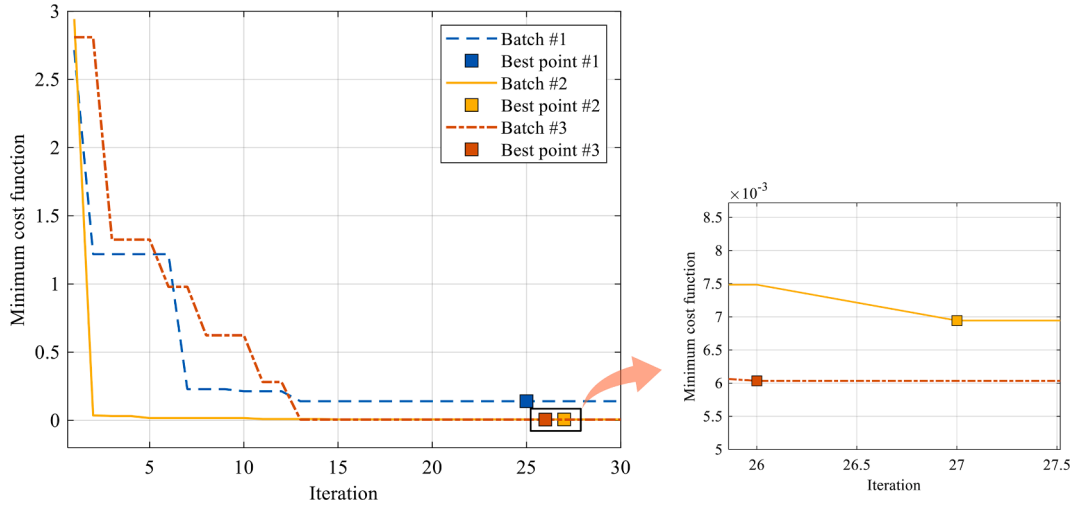


Fig. 10. Hyperparameter tuning processes of the ANN models trained by various data batches.

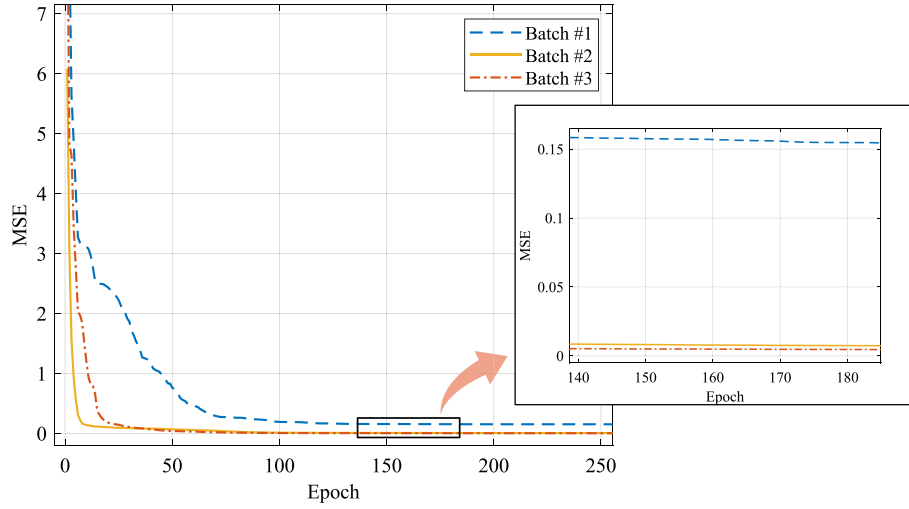


Fig. 11. Learning curves of the ANN models trained by various data batches.

of the ANN [19]. RMSE measures the dispersion of errors by calculating the square root of the average squared difference between predicted and true values.  $R^2$  is a metric that quantifies the proportion of the variability in the target variable that can be accounted for by the independent variables. It serves as an indicator of the model's goodness of fit to the observed data. The formulas for these metrics are presented as:

$$RMSE = \sqrt{\frac{1}{N} \sum_{j=1}^N (\hat{y}_j - y_j)^2} \quad (16)$$

$$R^2 = 1 - \frac{RMSE^2}{\sum_{j=1}^N (\bar{\hat{y}} - y_j)^2} \quad (17)$$

where  $\hat{y}_j$  and  $y_j$  are the predicted and actual values, respectively, and  $\bar{\hat{y}}$

represents the average of the predicted values.

Table 5 shows the values of the evaluation metrics for the training, validation, and test datasets. In order to check the influence of training data size on model predictions, the performance of the ANNs was also checked in relation to all of the data represented by the third batch as well. The analysis of data in Table 5 reveals a positive correlation between the amount of train data size and the prediction performances of the ANN models. Furthermore, a close agreement between the metric values obtained from the training and test sets within each batch confirms that the models can well fit into the new dataset, which comprises 20 % of the batch. However, a considerable difference appears when comparing the prediction results of the training set in the first batch and the prediction results of all data. This implies that the ANN model trained on the first data batch does not generalise well to the entire data

**Table 5**  
Values of evaluation metrics for ANNs.

| Dataset  | Training set |       | Validation set |       | Test set |       | All data (3rd batch) |       |
|----------|--------------|-------|----------------|-------|----------|-------|----------------------|-------|
|          | RMSE         | $R^2$ | RMSE           | $R^2$ | RMSE     | $R^2$ | RMSE                 | $R^2$ |
| Batch #1 | 0.234        | 0.976 | 0.211          | 0.988 | 0.309    | 0.976 | 0.655                | 0.941 |
| Batch #2 | 0.062        | 0.999 | 0.061          | 0.999 | 0.089    | 0.998 | 0.076                | 0.998 |
| Batch #3 | 0.044        | 0.999 | 0.044          | 0.999 | 0.080    | 0.999 | 0.054                | 0.999 |

space, suggesting that ANN's performance is sensitive to the variations present in the larger dataset that were not sufficiently captured during training with the smaller batch.

In addition, to visualise the accuracy of the prediction models, the correlations between the FE estimations and the ANN predictions for all data are plotted in Fig. 12. This figure compares the transverse modulus estimations between ANNs with different training sizes and the FE simulations. Fig. 12a shows the ANN predictions trained by the first batch. It is seen that the predicted values are located within  $\pm 30\%$  of the FE estimations. A higher dispersion is seen in the correlation for modulus values beyond 6 GPa in Fig. 12a, indicating that batch 1 did not sufficiently cover the diverse range of geometrical, mechanical, and environmental information that exists in the broader dataset. Fig. 12b and c show the correlations between of the ANN predictions, trained using second and third batches, and the FE calculations, respectively. The findings of ANN predictions show a close agreement with the FE results, with deviations fallings within the ranges of  $\pm 20\%$  and  $\pm 10\%$ . This suggests that the accuracy of the predictions was improved by enlarging the training dataset to include a wide range of values for the input features present in the dataset.

### 7.3. Influence of microstructure characteristics on transverse modulus

The ANN developed based on the third data batch was used to examine the impact of microstructure geometric features, which were obtained by the PCA method, on the effective transverse modulus. It was accomplished by performing a feature importance analysis using the permutation technique. In this method, the drop in the ANN model's accuracy caused by permuting the values of an input feature indicates a metric to measure its importance [64]. For an influential feature,

permuting its values greatly decreases the model's accuracy. The steps of ranking features by permutation are explained in the following.

1. For each  $PC_i$ , the data points were permuted.
2. The ANN model error from the permuted values was estimated.
3. The difference between the error before and after permutation was obtained.
4. The mean  $\mu_i$  and standard deviation  $\sigma_i$  of the difference was obtained for each input parameter.
5. The importance for each feature was obtained as  $\mu_i/\sigma_i$ .

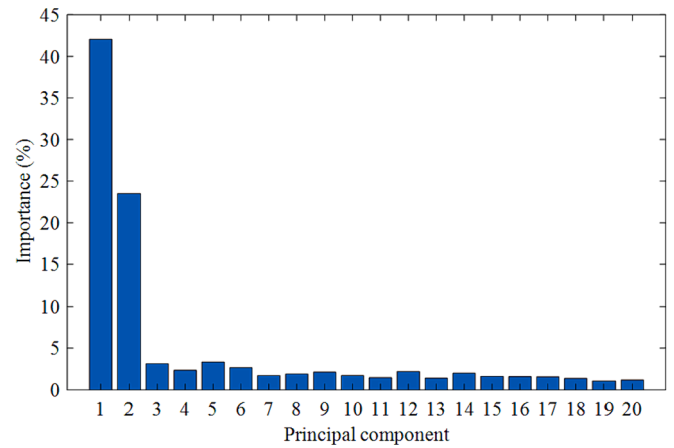


Fig. 13. Influence of PCs on Young's modulus prediction.

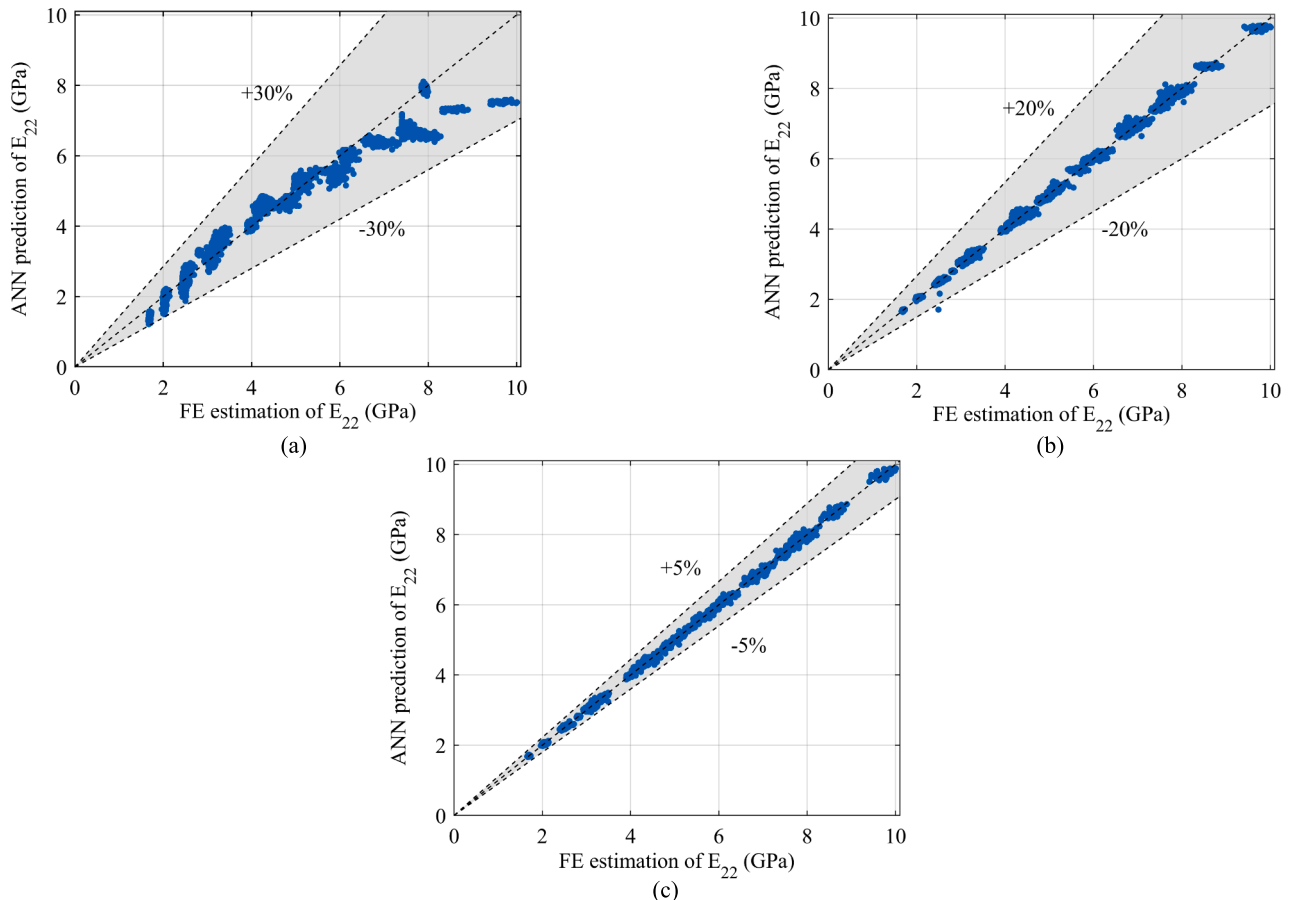


Fig. 12. Correlation between the FE simulations and the ANN predictions of  $E_{22}$ : (a) Batch #1, (b) Batch #2, (c) Batch #3.

Fig. 13 shows the importance of microstructure characteristics obtained by PCA. It is seen that  $PC_1$  and  $PC_2$  account for 42 % and 23 % of influence on the elastic modulus, respectively, whereas the remaining PCs each contribute less than 4 %. Additionally, the contour plots of the important components are provided in Fig. 14. Fig. 14a shows the plot of the first PC. It is seen that the highest intensity is located at the centre of the image. The  $PC_1$  provides information on the fibre volume fraction. The second PC plot in Fig. 14b accounts for an important geometrical feature, which is fibre size in relation to the RVE size [33]. Considering the high importance of the geometric features extracted by  $PC_1$  and  $PC_2$ , it is implied that the fibre volume fraction and the ratio of fibre to RVE size have a high influence on the transverse modulus of the UD composites.

The PC patterns of higher orders are generally more complex and encode further geometric features. It is seen in Fig. 14c and Fig. 14d that the 3rd and 4th PCs stand for the arrangement of fibres in the horizontal and vertical directions.  $PC_5$  in Fig. 14e is also important for predictions of the transverse modulus. However, the interpretation of the pattern is difficult to estimate.

#### 7.4. Influence of fibre volume fraction on transverse modulus

In order to study the influence of fibre volume fraction on the transverse modulus, a parametric study was conducted using the ANN trained by the third batch. In doing so, a new dataset was created by modifying the third batch and replacing the values of each input feature with their corresponding mode values, while keeping  $V_f$  with its original values. The created data were then imported into the ANN model to predict the effective transverse modulus with respect to changes in fibre volume fraction. Fig. 15 shows the change in the ANN predictions of  $E_{22}$  with different  $V_f$  values. For each fibre volume ratio, square and hexagonal periodic unit cells were generated and simulated for comparison. The ANN predictions of  $E_{22}$ , which were trained using RVEs with random fibre distributions, are shown to be constrained by the estimations derived from periodic square and hexagonal unit cells.

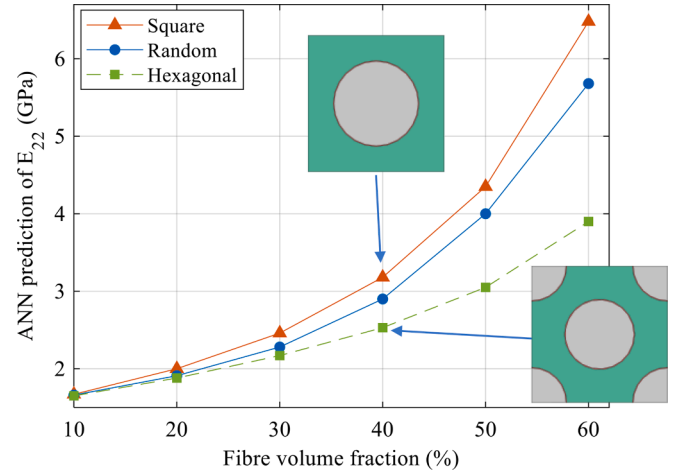


Fig. 15.  $E_{22}$  for various fibre volume fractions.

This offers a reasonable estimation of Young's modulus for the RVEs generated in this paper. It is shown in Fig. 15 that the discrepancies among the Young's modulus for models with different fibre arrangements increase with  $V_f$ . Regardless of volume fraction values, the distance between neighbouring fibres in square or hexagonal arrays remains constant. However, in a random distribution, it is possible that fibres are located very close to each other, especially in higher volume fractions, leading to high-stress values between them. To observe the influence of fibre distributions on the stress concentration at higher volume fractions, the local stress fields were plotted for two RVEs using FE simulations. The normalised stress contours for two micromodels with a 50 % volume fraction and different fibre distributions are displayed in Fig. 16. It is seen that stress concentrations are present in regions where the fibres are close together. Comparing the stress plots shows that the local stress values generated by the fibre arrangements in

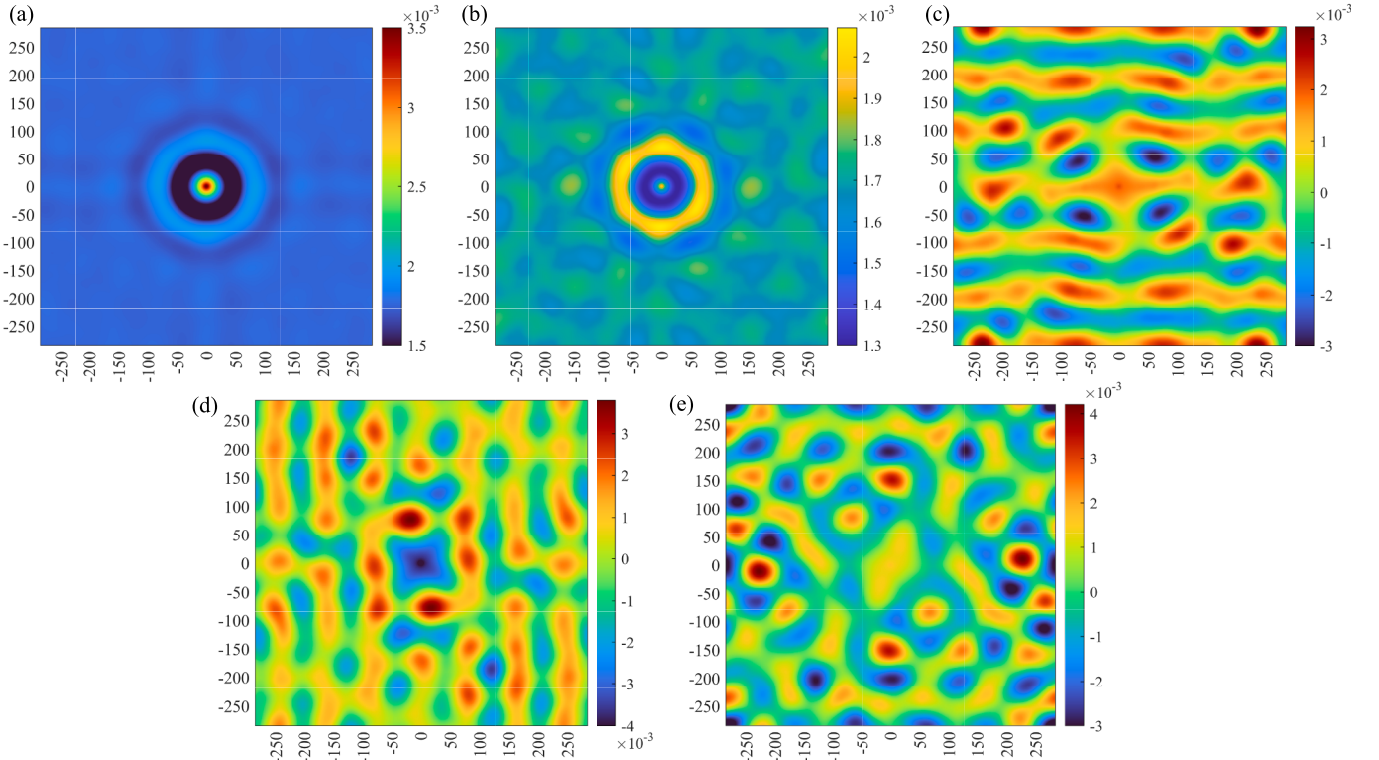


Fig. 14. Contour plots of different PCs: (a)  $PC_1$ , (b)  $PC_2$ , (c)  $PC_3$ , (d)  $PC_4$ , (e)  $PC_5$ .

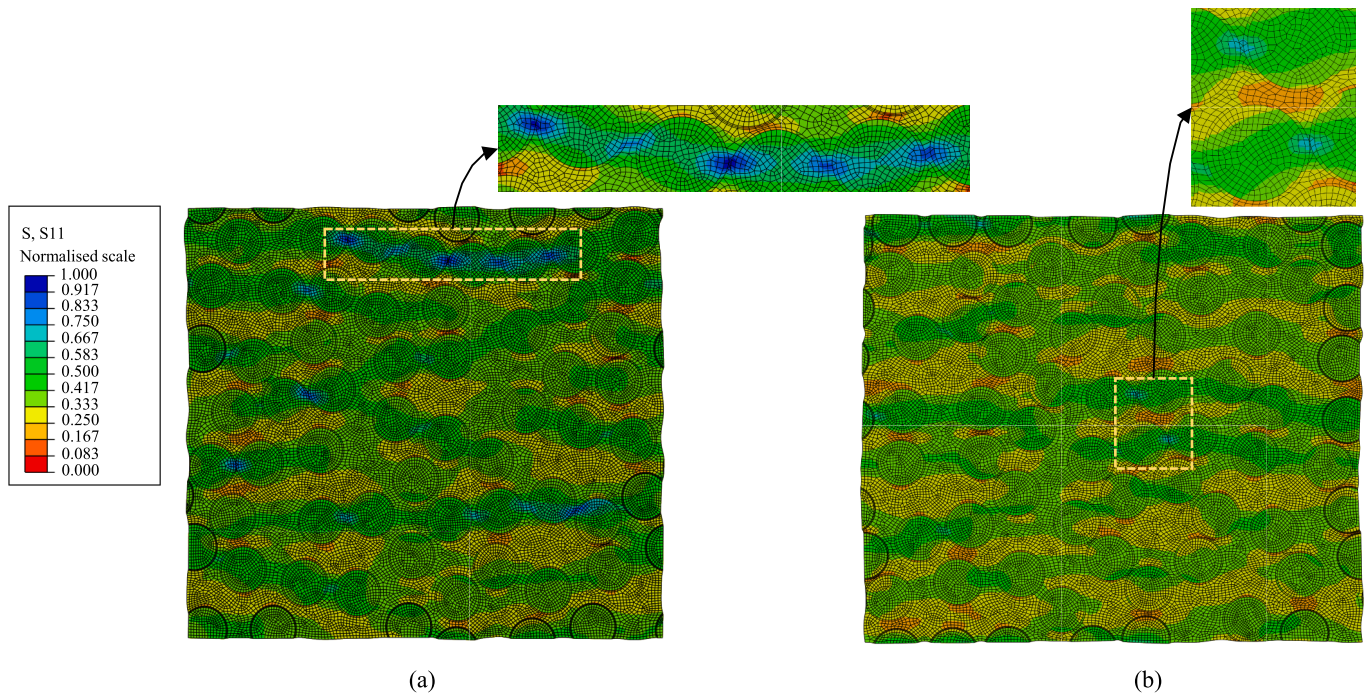


Fig. 16. Stress concentrations in RVEs with  $\nu_f = 50\%$ : (a) high local values ( $E_{22} = 4.9$  GPa), (b) low local values ( $E_{22} = 4.6$  GPa).

Fig. 16a are greater than those in Fig. 16b. Consequently, this disparity in stress distribution led to different transverse modulus values. This indicates that the values and locations of the stress concentrations, which are sources of damage, depend on the distribution of fibres in the matrix.

#### 7.5. Influence of interphase thickness on transverse modulus

Interphase is a key element in determining the crack propagation and, subsequently, the rupture mechanism of the composite [65]. The interphase can protect the fibres from environmental degradation [66]. Therefore, the overall performance of the composite is affected by the quality of the interphase area. The impact of the interphase region was addressed by a parametric study of the ANN predictions of transverse modulus over a range of interphase thickness and fibre volume fraction values. Fig. 17 depicts the change in the Young's modulus of the

microstructure as the interphase thickness ranges from 0 to 1000 nm. It should be noted that above this range, the Young's modulus of the matrix is equal to that of the bulk resin [50]. It is seen that the elastic modulus is inversely correlated to the interphase thickness, indicating that the rigidity of composites decreases with increasing interphase thickness. This is because the interphase used in the parametric study is softer than the matrix. Therefore, as the thickness of the interphase increases, the hard matrix is gradually replaced by the soft interphase, leading to a resultant decrease in the overall stiffness of the composite material. The data shown in Fig. 17 indicates that there is a correlation between the change in  $E_{22}$  values and the varying fibre volume fractions. This suggests that when the interphase thickness grows in composites with high fibre volume fraction, the influence of the soft interphase on the effective transverse properties becomes significant. This leads to a greater reduction in the stiffness when compared to the composite with a smaller volume fraction.

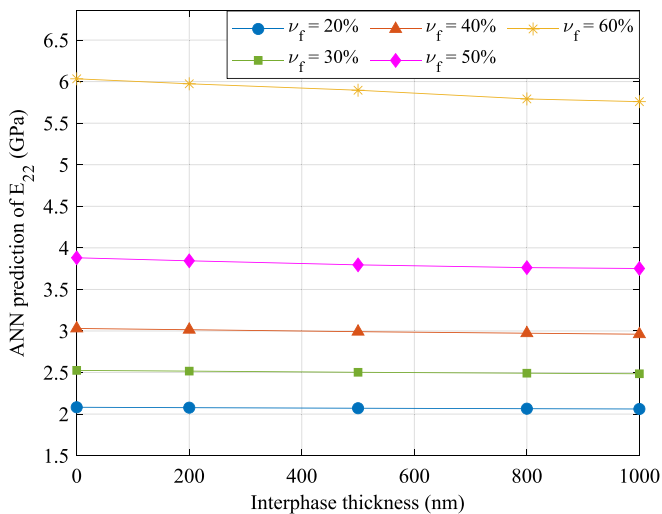


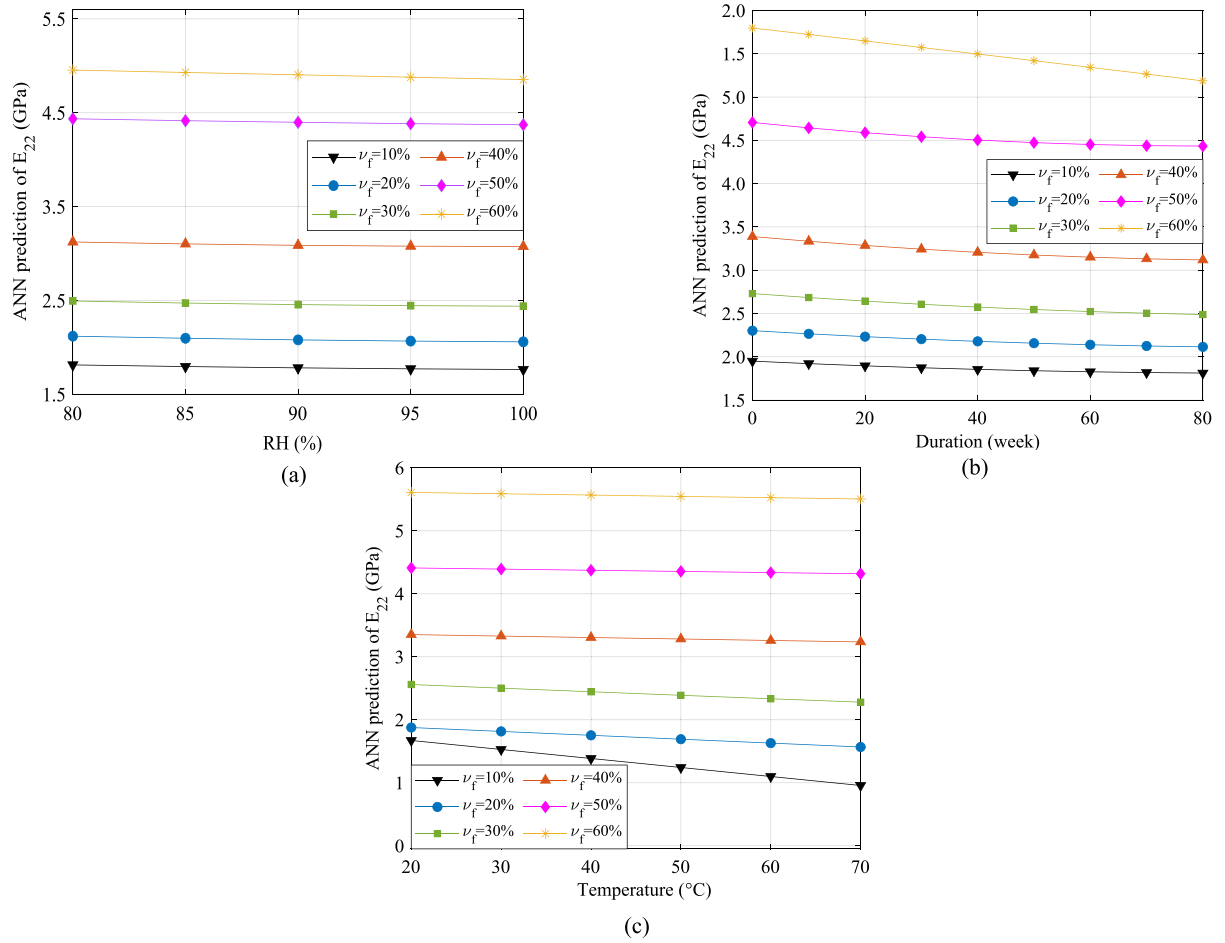
Fig. 17. Influence of interphase thickness on  $E_{22}$  for composites with various fibre volume fractions.

#### 7.6. Influence of humid environmental conditions on transverse modulus

In order to demonstrate the impact of the humid condition on the long-term transverse performance of the composite, a parametric study of environmental factors was conducted by the ANN model. In doing so, the third data batch was modified by replacing the input features with the corresponding mode values, while the data for RH, temperature, and duration of exposure parameters were kept at their original values. The modified data were then imported into the ANN model to predict  $E_{22}$  with respect to changes in environmental factors.

Fig. 18a shows the loss in  $E_{22}$  during 76 weeks of exposure to  $35^\circ\text{C}$  humidity with different levels of RH; Fig. 18b depicts the loss in transverse modulus with time for a humid condition with 80 % RH and  $35^\circ\text{C}$  humidity; and Fig. 18c shows the change in  $E_{22}$  of composite samples exposed to 80 % RH with varying temperature for 76 weeks. It is seen in these figures that as the RH, exposure duration, and temperature increase, the  $E_{22}$  values decrease, implying the adverse impact of the humid condition on the long-term transverse performance of the UD composite material.

Fig. 18a and b also show that the material degradation rate under the influence of humid conditions with varying RH and exposure duration



**Fig. 18.**  $E_{22}$  loss under humid condition: (a) impact of RH at 35 °C for 76 weeks, (b) impact of exposure duration under 80 % RH and 35 °C, (c) impact of temperature under 80 % RH for 76 weeks.

values is independent from the fibre volume fraction. However, it is inversely related to  $V_f$  when exposed to changing temperature values. This implies that the composite with a high ratio in the resin matrix volume is susceptible to degradation when exposed to hygrothermal ageing, which is related to the molecular relaxation and plasticisation in resin, due to which the stiffness of the material decreases [67].

## 8. Conclusions

In this study, an efficient method was developed to establish a structure-long-term property linkage by taking into account the microstructural characteristics and the mechanical and environmental properties of glass fibre/epoxy composites under humid ageing. A framework was developed to artificially generate periodic composite microstructure while accounting for the interphase region between fibre and matrix. A comparison between the numerical, experimental, and analytical approaches verified the validity of the FE simulations for predicting material degradation under a variety of environmental parameters.

A database establishment framework was developed to generate three batches of datasets with 1000, 2000, and 3000 sizes. For each generated RVE, TPC functions were applied to statistically measure the geometrical features of the microstructure. The PCA was performed to reduce the extracted 649,800 physical descriptors from each microstructure image to 20 PCs. It was seen that the first two PCs contained 98.5 % of the geometrical information. An ANN-based framework was developed by integrating Bayesian optimisation and 10-fold CV techniques to tune the model hyperparameters for optimal performance. In order to check the influence of training data size on predictions of

transverse modulus, ANN models were developed using the three data batches. The results indicated that there is an improvement in the performance of ANN as the RMSE decreases from 0.655 to 0.054 and the  $R^2$  value increases from 0.941 to 0.999 when the training size is increased from the first to the third batch.

A parametric study on the influence of microstructure characteristics revealed that the first five PCs contribute more than 65 % to the transverse performance of the UD composite while carrying information on volume fraction, size, and the arrangement of fibres in parallel and perpendicular directions to the load. Studying the influence of  $V_f$  on transverse modulus revealed a strong correlation between stiffness and fibre volume fraction of the composite material. A contribution of fibre distribution to the local stress was observed in composites with a 50 % fibre volume fraction, where stress concentrations occurred in regions with the fibres located closely together. The study on the influence of interphase region by ANN showed an inverse correlation between the thickness values varying from 0 to 1000 nm and the transverse modulus in materials having a softer interphase region than the matrix. Additionally, it was observed that the influence of thickness on elastic modulus increases with fibre volume fraction.

A parametric study of the humid condition characteristics by ANN revealed the adverse impact of RH, temperature, and humidity duration on the long-term performance of the UD composite material. It was seen that the material degradation rate is independent of the fibre volume fraction under the influence of different RH and exposure duration values. However, the degradation rate in transverse properties is adversely related to the fibre volume ratio under varying temperatures, implying the susceptibility of the resin matrix to hygrothermal ageing.

## CRediT authorship contribution statement

**K. Aghabalaee Baghaei:** Conceptualization, Data curation, Formal analysis, Investigation, Software, Validation, Visualization, Writing – original draft. **S.A. Hadigheh:** Conceptualization, Data curation, Funding acquisition, Investigation, Methodology, Project administration, Resources, Supervision, Validation, Writing – review & editing.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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